

33.1 Features

- Supports both standard JTAG and ARM Serial Wire Debug modes.
- Direct debug access to all memories, registers, and peripherals.
- No target resources are required for the debugging session.
- Trace port provides CPU instruction trace capability. Output can be via a 4-bit trace data port, or Serial Wire Viewer.
- Eight Breakpoints. Six instruction breakpoints that can also be used to remap instruction addresses for code patches. Two data comparators that can be used to remap addresses for patches to literal values.
- Four data Watchpoints that can also be used as trace triggers.
- Instrumentation Trace Macrocell allows additional software controlled trace.

33.2 Introduction

Debug and trace functions are integrated into the ARM Cortex-M3. Serial wire debug and trace functions are supported in addition to a standard JTAG debug and parallel trace functions. The ARM Cortex-M3 is configured to support up to eight breakpoints and four watchpoints.

33.3 Description

Debugging with the LPC176x/5x defaults to JTAG. Once in the JTAG debug mode, the debug tool can switch to Serial Wire Debug mode.

Trace can be done using either a 4-bit parallel interface or the Serial Wire Output. When the Serial Wire Output is used, less data can be traced, but it uses no application related pins. Parallel trace has a greater bandwidth, but uses 5 functional pins that may be needed in the application. Note that the trace function available for the Cortex-M3 is functionally very different than the trace that was available for previous ARM7 based devices, using only 5 pins instead of 10.

33.4 Pin Description

The tables below indicate the various pin functions related to debug and trace. Some of these functions share pins with other functions which therefore may not be used at the same time. Use of the JTAG port excludes use of Serial Wire Debug and Serial Wire Output. Use of the parallel trace requires 5 pins that may be part of the user application, limiting debug possibilities for those features. Trace using the Serial Wire Output does not have this limitation, but has a limited bandwidth.

Table 609. JTAG pin description

Pin Name	Type	Description
TCK	Input	JTAG Test Clock. This pin is the clock for debug logic when in the JTAG debug mode.
TMS	Input	JTAG Test Mode Select. The TMS pin selects the next state in the TAP state machine.
TDI	Input	JTAG Test Data In. This is the serial data input for the shift register.
TDO	Output	JTAG Test Data Output. This is the serial data output from the shift register. Data is shifted out of the device on the negative edge of the TCK signal.
$\overline{\text{TRST}}$	Input	JTAG Test Reset. The $\overline{\text{TRST}}$ pin can be used to reset the test logic within the debug logic.
RTCK	Output	JTAG Returned Test Clock. This is an extra signal added to the JTAG port, and is included for backward pin compatibility with LPC23xx series devices that share the same pinout as this device. RTCK is not normally used with the Cortex-M3. For designs based on ARM7TDMI-S processor core, this signal could be used by external JTAG host interface logic to maintain synchronization with targets having a slow or varying clock frequency. For details refer to "Multi-ICE System Design considerations Application Note 72 (ARM DAI 0072A)".

Table 610. Serial Wire Debug pin description

Pin Name	Type	Description
SWDCLK	Input	Serial Wire Clock. This pin is the clock for debug logic when in the Serial Wire Debug mode.
SWDIO	Input / Output	Serial wire debug data input/output. The SWDIO pin is used by an external debug tool to communicate with and control the Cortex-M3 CPU.
SWO	Output	Serial Wire Output. The SWO pin can deliver Serial Wire Viewer data only. Serial Wire Viewer data includes ITM and DWT trace data.

Table 611. Parallel Trace pin description

Pin Name	Type	Description
TRACECLK	Input	Trace Clock. This pin provides the sample clock for trace data on the TRACEDATA pins when tracing is enabled by an external debug tool.
TRACEDATA[3:0]	Output	Trace Data bits 3 to 0. These pins provide ETM trace data when tracing is enabled by an external debug tool. The debug tool can then interpret the compressed information and make it available to the user.

33.5 Debug Notes

Important: The user should be aware of certain limitations during debugging. The most important is that, due to limitations of the Cortex-M3 integration, the LPC176x/5x cannot wake up in the usual manner from Deep Sleep and Power-down modes. It is recommended not to use these modes during debug.

Another issue is that debug mode changes the way in which reduced power modes are handled by the Cortex-M3 CPU. This causes power modes at the device level to be different from normal modes operation. These differences mean that power measurements should not be made while debugging, the results will be higher than during normal operation in an application.

During a debugging session, the System Tick Timer and the Repetitive Interrupt Timers are automatically stopped whenever the CPU is stopped. Other peripherals are not affected. If the Repetitive Interrupt Timer is configured such that its PCLK rate is lower than the CPU clock rate, the RIT may not increment predictably during some debug operations, such as single stepping.

Debugging is disabled if code read protection is enabled.

The bootloader uses the SRAM during execution. When a system reset is issued to the core and peripherals during a debug session, SRAM 0x10000120 will be used by the bootloader. The boot code uses a complete word ("0x10000120...0x10000123" or "32-bit SRAM word 0x10000120").

33.6 Debug memory re-mapping

Following chip reset, a portion of the Boot ROM is mapped to address 0 so that it will be automatically executed. The Boot ROM switches the map to point to Flash memory prior to user code being executed. In this way a user normally does not need to know that this re-mapping occurs.

However, when a debugger halts CPU execution immediately following reset, the Boot ROM is still mapped to address 0 and can cause confusion. Ideally, the debugger should correct the mapping automatically in this case, so that a user does not need to deal with it.

33.6.1 Memory Mapping Control register (MEMMAP - 0x400F C040)

The MEMMAP register allows switch the mapping of the bottom of memory, including default reset and interrupt vectors, between the Boot ROM and the bottom of on-chip Flash memory.

Table 612. Memory Mapping Control register (MEMMAP - 0x400F C040) bit description

Bit	Symbol	Value	Description	Reset value
0	MAP		Memory map control.	0
		0	Boot mode. A portion of the Boot ROM is mapped to address 0.	
		1	User mode. The on-chip Flash memory is mapped to address 0.	
31:1	-		Reserved. The value read from a reserved bit is not defined.	NA

33.7 JTAG TAP Identification

The JTAG TAP controller contains device ID that can be used by debugging software to identify the general type of device. More detailed device information is available through ISP/IAP commands (see [Section 32.7](#) and [Section 32.8](#)). For the LPC176x/5x family, this ID value is 0x4BA0 0477.